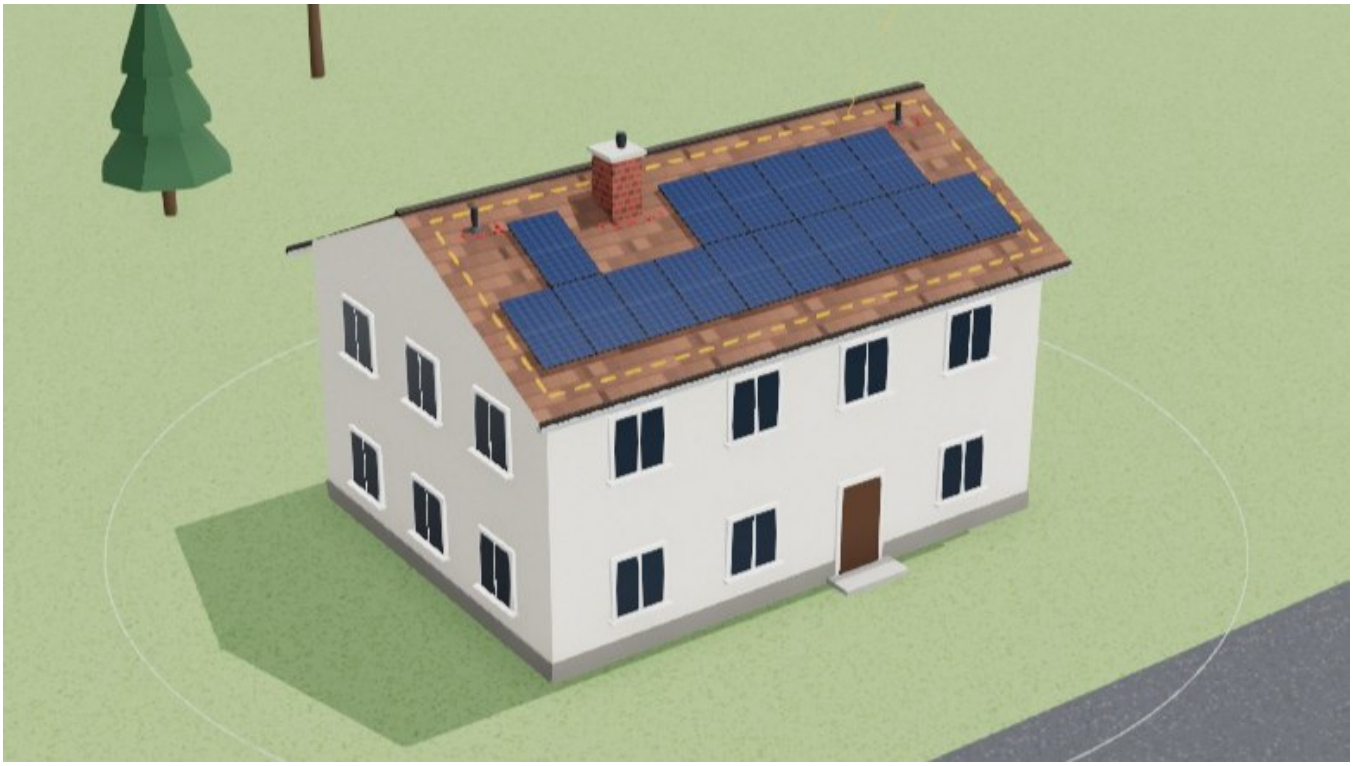
SOLAR PV PROPOSAL

6.9 kWp house · gable roof

New Delhi · 22 October 2026

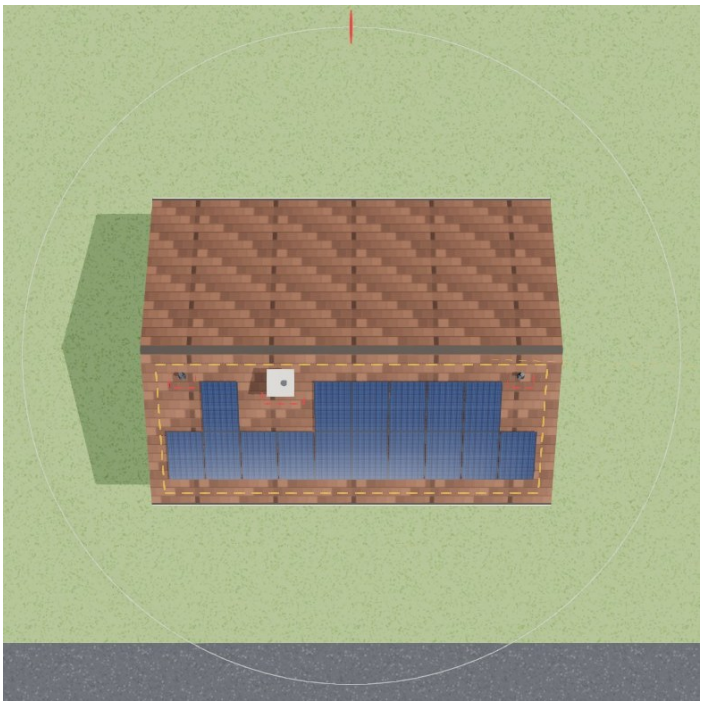


16 modules	**6.9 kWp** DC capacity	**9,758 kWh** per year (P50)	**1,418 kWh/kWp** specific yield
INR 3,44,000 installed cost	**INR 78,067** saved per year	**4.4 yr** simple payback	**6.9 t** CO2 avoided / yr

16 x 430 W modules (31 m²) are laid out over 1 array, facing 180° S. Energy is modelled from ten years of measured NASA POWER weather for this location (2016-2025), and the system is expected to deliver 9,758 kWh in a typical year and at least 9,451 kWh in nine years out of ten (P90). A shading study of every module across the year puts shading losses at 0.1 %.

System design

New Delhi



Plan view - amber: usable area after setbacks, red: keep-out around obstacles

Site	House · gable roof
Module	430 W residential
	108 half-cell TOPCon · 1722 x 1134 mm
Orientation	Portrait
DC capacity	6.9 kWp
AC inverter	5.7 kW (DC:AC 1.2)
Module area	31 m² (22 % of surface)
Weight	0.34 t · 13.8 kg/m² dead load
Setbacks	edge 0.45 m · ridge 0.45 m
Obstacles	1 chimney, 2 vents

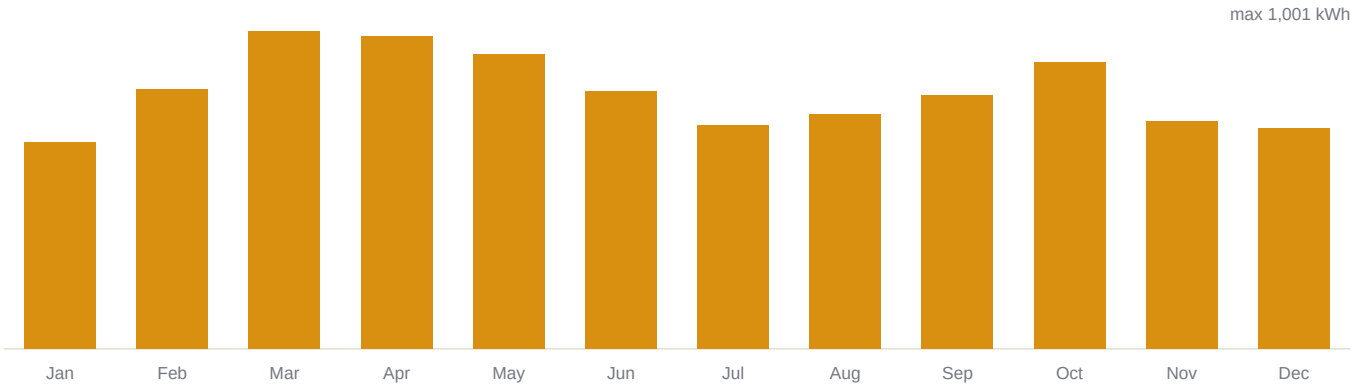
ARRAYS

	Plane	Tilt	Faces	Modules	kWp	Shade	kWh / yr
A	Front roof plane	26°	180° S	16	6.9	0.1 %	9,758

Energy yield

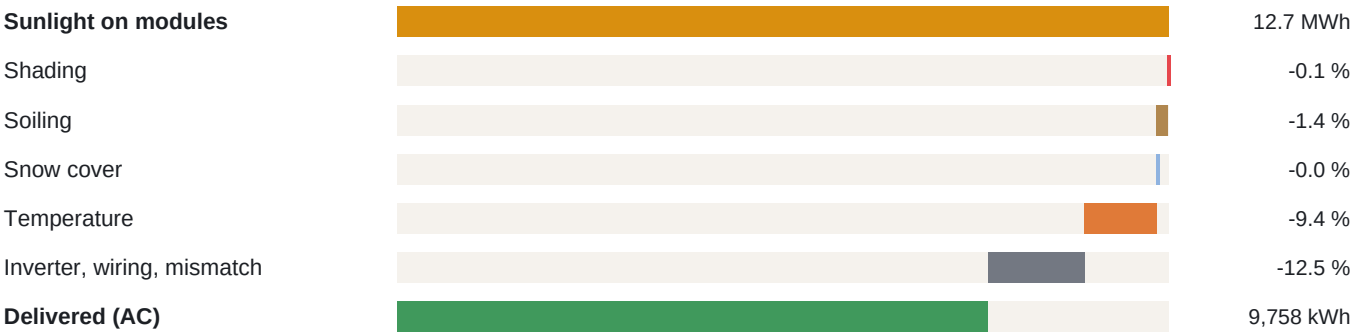
NASA POWER 2016-2025

ENERGY BY MONTH (KWH)

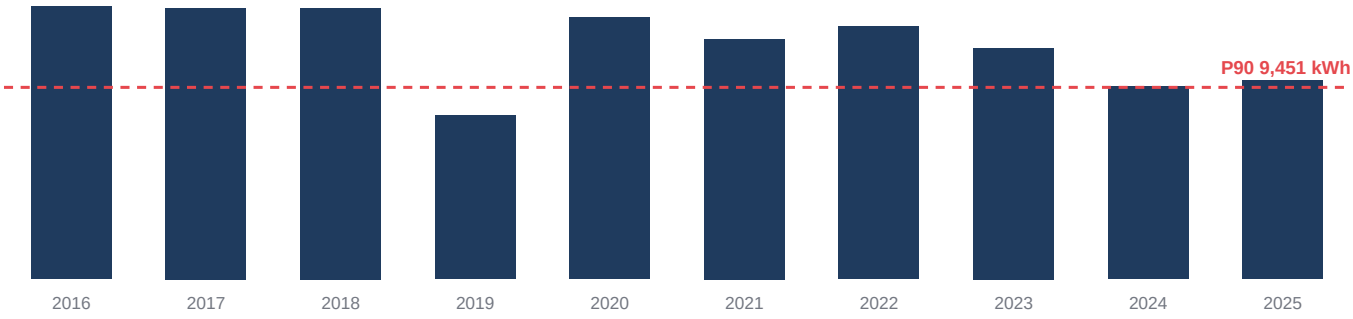


1,418	16.2 %	76.6 %	0.1 %
kWh/kWp per year	capacity utilisation (CUF)	performance ratio	shading loss

WHERE THE SUNLIGHT GOES (PER YEAR)



ENERGY YEAR BY YEAR, WITH P90

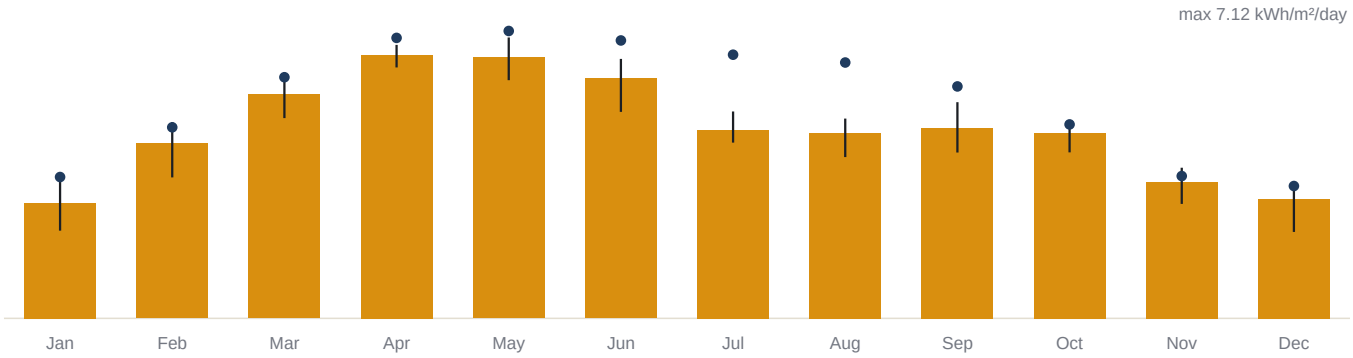


Site climate

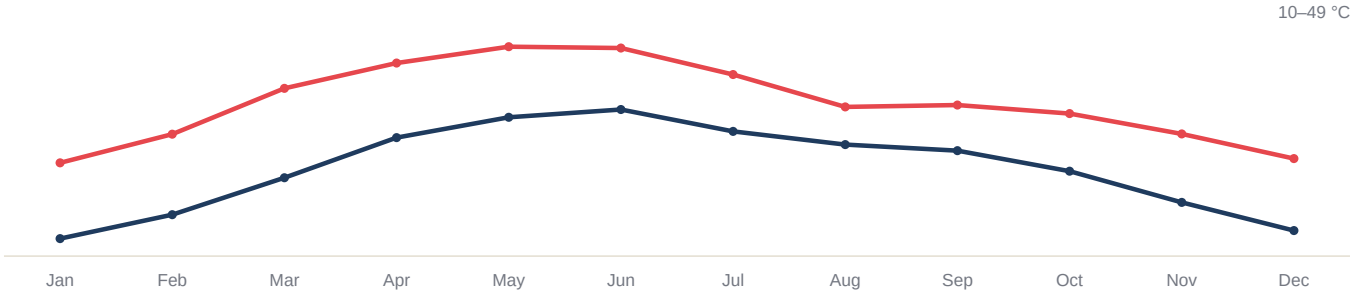
New Delhi · 2016-2025

1,719	4.71 h	25.3 °C	794 mm
kWh/m² per year (GHI)	peak sun hours / day	mean air temperature	rain per year

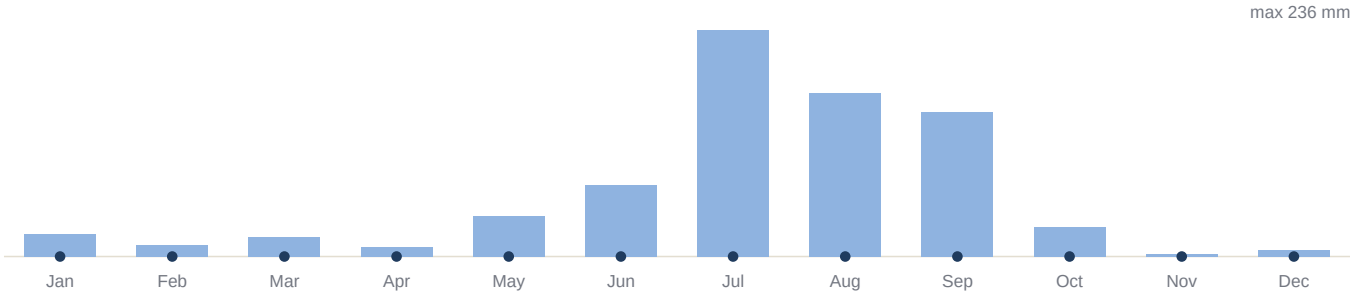
SOLAR IRRADIATION, KWH/M² PER DAY - BAR: MEAN, LINE: 10-YR RANGE, DOT: CLEAR SKY



TEMPERATURE - RED: DAILY MAX, NAVY: MEAN (°C)



RAIN (BARS) AND SNOWFALL WATER-EQUIVALENT (DOTS), MM PER MONTH



Dust costs about 1.4 % a year with cleaning every 30 days. Snow does not settle here. Heat costs 9.4 % through cell temperature.

Financials

INR · 50.00 per Wp · 8.00 per kWh

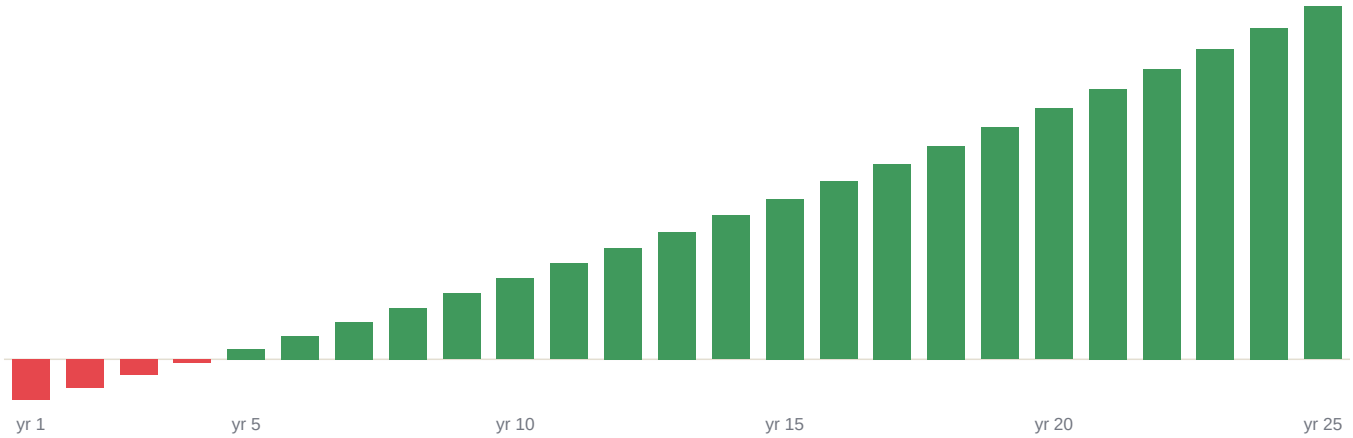
INR 3,44,000
installed cost

INR 78,067
first-year savings

4.4 yr
simple payback

INR 26,61,414
25-year savings

CUMULATIVE CASH POSITION OVER 25 YEARS (TARIFF +3 %/YR, OUTPUT -0.5 %/YR)



229.9 MWh
25-year energy

6.9 t
CO2 avoided per year

163 t
CO2 avoided, 25 yr

INR 1.50
cost per kWh (no O&M, no finance)

ASSUMPTIONS & SOURCES

- Modules: 430 W residential, 108 half-cell TOPCon · 1722 x 1134 mm. Inverter sized at DC:AC 1.2. Linear degradation 0.5 %/yr.

- Weather: NASA POWER monthly means (CERES / MERRA-2) for the last ten complete years. Horizontal irradiation split into beam and diffuse (Collares-Pereira & Rabl) and transposed per plane (isotropic sky, albedo 0.2). Temperature loss -0.35 %/°C above 25 °C cell; 14 % system losses.

- Soiling: dust build-up by rainfall class, cleared by rain events and the cleaning interval. Snow: covered days from snowfall, tilt and temperature.

- Shading: every module centre ray-cast to the sun every half hour on a mid-month day, against trees, rows, roof planes and obstacles; beam light only.

- Rows and setbacks: rows spaced for no shade at winter-solstice noon; roof setbacks per IFC/IRC (18" ridge, 36" pathways) unless changed.

- Data: NASA POWER (power.larc.nasa.gov); OpenStreetMap contributors (ODbL); Open-Meteo geocoding.

This is a concept proposal, not an engineering design. Structural capacity, electrical design, local codes, permits, grid connection and final pricing must be confirmed by a site survey and a qualified installer.